1)

a. Assumptions

1. OrderNo does not appear on the form, but the combination of customerNo + date + time uniquely identifies an order
2. customerNo (e.g., HG54587) uniquely identifies a customer
3. partNo uniquely identifies a part
4. An employee is identified by their name (employeeName)
5. Cage code is an attribute of a part, not of the order
6. unitPrice is a property of the part, not of a specific order line
7. quantityOrdered is specific to a particular order line

B

Step 1: 1NF.

The original chart is UNF. The form shows one order with four line items — the part-related attributes (partNo, partName, type, cageCode, quantityOrdered, unitPrice) form a repeating group.
To change into 1NF I eliminated the repeating group by making partNo part of the composite primary key. Each row is now uniquely identified by {customerNo, date, time, partNo}. All attributes are now atomic.
PK: {customerNo, date, time, partNo}

customerNo	customerName	customerType	**date**	**time**	employeeName	**partNo**	partName	type	cageCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10654	Float Control	Plumbing	G413	4	12
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

Step 2: Second Normal Form (2NF)
Fix: Remove all partial dependencies by decomposing into four separate relations. Each non-key attribute now depends on the whole key of its relation. In each relation, all non-key attributes depend directly and only on the primary key — there are no transitive dependencies. Therefore 2NF = 3NF, with foreign keys

C. D.

Customer— PK: {customerNo}

customerNo	customerName	customerType
HG54587	Jeff Peterson	Consumer

Order — PK: {customerNo, date, time}

customerNo	**date**	**time**	employeeName
HG54587	7/1/2024	10:30am	D. Harrison

FK: customerNo → Customer

Part — PK: {partNo}

partNo	partName	type	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

OrderLine — PK: {customerNo, date, time, partNo}

customerNo	***date***	***time***	***partNo***	quantityOrdered
HG54587	7/1/2024	10:30am	10654	4
HG54587	7/1/2024	10:30am	10456	3
HG54587	7/1/2024	10:30am	10776	7
HG54587	7/1/2024	10:30am	10657	5

FK: customerNo, date, time → Order | FK: partNo → Part

2)
a. Assumptions
1. staffNo uniquely identifies a therapist and determines therapistName.
2. patNo uniquely identifies a patient and determines patName.
3. A therapist works at one specific branch on any given day
4. An appointment is uniquely identified by {staffNo, patNo, date, time}.
5. A patient can have multiple appointments per day with different therapists.
6. branchNo is where the appointment takes place, determined by which branch the therapist works at on that date.

Step 1: 1NF
The table is already atomic and no multi-valued attributes exist. We flatten the repeating group by identifying
a composite PK. Each row is uniquely identified by {staffNo, patNo, date, time}.
PK: {staffNo, patNo, date, time}

staffNo	therapistName	**patNo**	patName	**date**	**time**	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

Partial dependencies exist:
• therapistName depends only on staffNo
• patName depends only on patNo
• branchNo depends only on {staffNo, date}

Step 2: 2NF

Therapist — PK: {staffNo}

staffNo	therapistName

S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

Patient — PK: {patNo}

patNo	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

TherapistBranch — PK: {staffNo, date}

staffNo	**date**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

FK: staffNo → Therapist

Appointment – PK: {staffNo, patNo, date}

staffNo	***patNo***	***date***	**time**
S1011	P100	9/12/2022	10:00
S1011	P105	9/12/2022	12:00
S1024	P108	9/12/2022	10:00
S1024	P108	9/14/2022	14:00
S1032	P105	9/14/2022	16:30
S1032	P110	9/15/2022	18:00

FK: staffNo, date → TherapistBranch | FK: patNo → Patient

3)
a. Assumptions
1. eNo uniquely identifies an employee and determines eName.
2. contractNo uniquely identifies a contract and determines eventNo and eventLoc.
3. Each contract applies to only one event (stated in the problem).
4. eventNo determines eventLoc — an event has one location.
5. hours worked by an employee on a contract depends on both eNo and contractNo.
6. There may be multiple contracts for the same event.

B.
step 1: 1NF

eNo	**contractNo**	**hours**	**eName**	**eventNo**	**eventLoc**
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

Step 2: 2NF

Removed Partial dependencies: eName depends only on eNo, eventNo, eventLoc depend only on contractNo, and Only hours depends on the full key {eNo, contractNo}

Employee — PK: {eNo}

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

Contract — PK: {contractNo}

contractNo	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers

C1026	H25	Queens

EmployeeContract — PK: {eNo, contractNo}

eNo	**contractNo**	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

Step 3: 3NF

In the Contract relation, eventLoc depends on eventNo, which depends on contractNo. This is a transitive dependency (contractNo → eventNo → eventLoc). Remove the transitive dependency by splitting the Contract relation into Contract and Event.

Event — PK: {eventNo}

eventNo	eventLoc
H25	Queens
H4	Yonkers

Contract — PK: {contractNo}

contractNo	*eventNo*
C1024	H25
C1025	H4
C1026	H25

FK: eventNo → Event

Employee — PK: {eNo}

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

EmployeeContract — PK: {eNo, contractNo}

eNo	***contractNo***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

FK: eNo → Employee | FK: contractNo → Contract